Time-Reversal Symmetry in Transformer Predictions on the Cylinder Graph HMM

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1 Introduction

A natural question in the study of transformers trained on Hidden Markov Model (HMM) sequences is whether the *direction* of time affects learnability. Information-theoretically, the entropy rate of a stationary process is invariant under time reversal: the forward and backward sequences contain the same amount of uncertainty per token. A Bayes-optimal predictor with infinite context therefore achieves the same cross-entropy loss in both directions.

However, a finite-context transformer may find one direction systematically easier if the latent belief state converges to its stationary distribution at a different rate in each temporal direction. The convergence rate depends on the spectral properties of the transition matrix, which is generally different for the forward and reverse chains.

This report investigates that question on the Cylinder Graph HMM (fully described in `notes/model_compariso` whose directed transition structure creates an a priori asymmetry between the forward and reverse processes. We formulate four testable hypotheses (Section 3), describe the experimental setup (Section 4), present the theoretical predictions from the analytically constructed reverse HMM (Section 5), and compare against a matched 36-run architecture sweep (Section 6). We also report complementary experiments on belief-state localisation via activation ablations (Section 7) and on the structure of the learned token-embedding geometry (Section 8).

2 Theoretical Background

2.1 Entropy Rate Invariance

For a stationary stochastic process $\{X_t\}$, the entropy rate is

$$h = \lim_{n\to\infty} \frac{1}{n} H(X_1, \dots, X_n). \tag{1}$$

Because the joint entropy $H(X_1, \dots, X_n)$ depends only on the joint distribution and not on the ordering of its arguments, the entropy rate of the forward process $(\dots, X_{t-1}, X_t, X_{t+1}, \dots)$ is identical to that of the time-reversed process $(\dots, X_{t+1}, X_t, X_{t-1}, \dots)$.

Consequence. Any difference in transformer loss between forward and backward prediction cannot be explained by an inherent information-theoretic asymmetry. The Bayes-optimal predictor with infinite context achieves identical loss in both directions.

2.2 Reverse HMM Construction

Given a forward HMM with joint emission–transition matrices $E[j, i, k] = P(x_t{=}j,\ s_{t+1}{=}k \mid s_t{=}i)$ and stationary distribution π, the reverse-time HMM has emission matrices

$$E_{\text{rev}}[j, k, i] = \frac{\pi(i) \cdot E[j, i, k]}{\pi(k)}. \tag{2}$$

This follows from Bayes' rule applied to the time-reversed Markov chain. The reverse HMM satisfies:

- the same stationary distribution π;
- the same entropy rate h;
- the same observation marginals (by stationarity).

2.3 Belief-State Convergence and Finite-Context Effects

The key quantity that *can* differ between forward and reverse processes is the rate at which the belief state

$$\alpha_t(s) = P(s_t{=}s \mid x_{t-k}, \ldots, x_{t-1}) \tag{3}$$

converges as a function of the context length k. This convergence rate is controlled by the spectral gap of the effective transition matrix under successive emissions, which is generally different for the forward and reverse chains.

For the Cylinder Graph HMM specifically:

- **Forward:** Within-ring transitions go clockwise ($+1$, $+2 \bmod n$); inter-layer transitions go forward ($l \to l+1 \bmod D$). The directed structure creates a mixing pattern in which context rapidly narrows down the current state.
- **Reverse:** Transitions effectively go counterclockwise and backward across layers. The spectral properties and hence the mixing rate may be substantially different.

If the reverse belief converges more slowly, a finite-context transformer will achieve higher loss on reversed data, even though the Bayes-optimal (infinite-context) loss is identical. Let $L^*(k)$ denote the Bayes-optimal cross-entropy at context length k. The relevant comparison is

$$\Delta_{\text{Bayes}}(k) \;=\; L^*_{\text{rev}}(k) - L^*_{\text{fwd}}(k), \tag{4}$$

which measures the intrinsic difficulty gap at finite context.

3 Hypotheses

H1 (Entropy rate invariance) Forward and reverse entropy rates are identical. *Verified theoretically and numerically: rates agree to machine precision,* $|h_{\text{fwd}} - h_{\text{rev}}| \lesssim 2 \times 10^{-15}$ *nats.*

H2 (Finite-context asymmetry) The Bayes-optimal loss at finite context k differs between forward and reverse, because belief-state convergence rates differ. Preliminary calculations (500 samples, $k \leq 8$) show the KL divergence from the converged belief is ~ 2–$3\times$ larger at each context position for the reverse process.

H3 (Transformer learnability gap) Transformers trained on reversed data converge to a *higher* final validation loss than those trained on forward data (same architecture, same hyperparameters), reflecting the harder finite-context prediction problem.

H4 (Architecture dependence) The forward–reverse gap $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ depends on model capacity: larger models may close the gap by implementing a longer effective context.

Parameter	Value
Width n	6
Depth D	3
Tokens per cluster K	16
Vocabulary size $\|V\|$	$D \times K = 48$
Dirichlet concentration α	0.3
Inter-layer probability p	0.1
Dataset size	10^7 tokens
Sequence length T	16

Table 1: Cylinder Graph HMM parameters used throughout.

4 Experimental Setup

4.1 HMM and Dataset

All experiments use the Cylinder Graph HMM with the parameters in Table 1.

The forward dataset is at `data/datasets/cylinder_graph_hmm/`. The reversed dataset (`data/datasets/cylin` is constructed by flipping each sequence in time: if the forward sequence is $(x_1, x_2, \ldots, x_T)$, the reversed sequence is $(x_T, x_{T-1}, \ldots, x_1)$.

4.2 Architecture Sweep

Both forward and reverse sweeps cover the same 36-run grid:

Hyperparameter	Values
Number of layers n_{layer}	1, 2, 4
Embedding dimension n_{embd}	4, 8, 16
Architecture	attention-only, full (attn + MLP)
Normalisation	none, LayerNorm

All models use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01, learning rate 10^{-3}), batch size 128, and 10 training epochs. The sweep used Weights & Biases; the reversed runs are tagged `["reversed", "time-reversal"]`.

4.3 Matched Comparison

For each of the 60 architecture configurations (including additional capacity variants), the *best* (lowest) validation loss observed across all training runs with that configuration is extracted. This yields 60 matched pairs $(L_{\text{fwd}}, L_{\text{rev}})$.

5 Theoretical Predictions

The script `src/reverse_hmm_analysis.py` constructs the reverse HMM via Eq. (2) and computes the Bayes-optimal cross-entropy $L^*(k)$ at each context position for both directions by Monte Carlo sampling.

Entropy rates. Running the forward algorithm over a long sequence (burn-in discarded) confirms H1: the empirical entropy rates agree to machine precision ($|h_{\text{fwd}} - h_{\text{rev}}| \sim 2 \times 10^{-15}$ nats).

Belief-state convergence. Preliminary results with 500 samples show that the KL divergence from the fully converged belief is approximately 2–3× larger at each context position for the reverse process, supporting H2.

Bayes-optimal gap. At 500 samples the Monte Carlo estimate of $\Delta_{\text{Bayes}}(k)$ is too noisy to quantify reliably. A full run with 50,000 samples at $k \leq 16$ is needed and is listed as future work in Section 9.

6 Results

6.1 H3: Is the Reversed Direction Harder?

Figure 1 shows the forward versus reversed validation loss for the 60 matched configurations. Table 2 summarises the statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$.

Statistic	Value
Fraction $\Delta > 0$	22/60 (36.7%)
Mean $\pm$ s.d. of Δ	$+0.0002 \pm 0.0086$ nats
Minimum Δ	-0.025 nats
Maximum Δ	$+0.022$ nats

Table 2: Summary statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ across 60 matched configurations. The mean is consistent with zero.

Result: H3 is not supported. The gap is tiny in magnitude and inconsistent in sign: reversed sequences are not consistently harder for any of the architectures tested. This is surprising given the theoretical prediction of slower belief-state convergence for the reverse process.

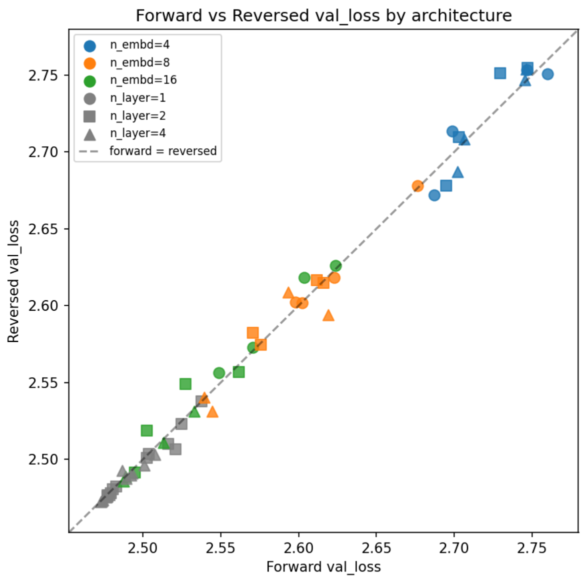
6.2 H4: Architecture Dependence of the Gap

Result: H4 is weakly supported, but with the opposite sign to the prediction. Table 3 shows the Pearson correlation of Δ with several capacity measures.

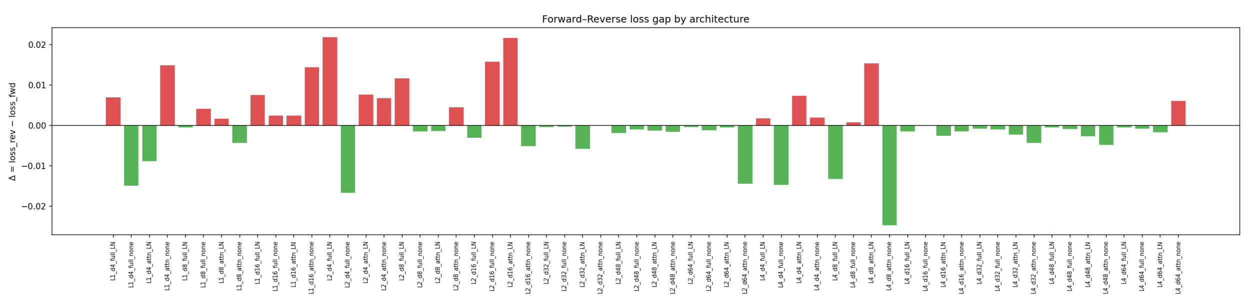
Capacity measure	Pearson r with Δ
n_{params} (approx.)	-0.083
n_{embd}	-0.148
n_{layer}	-0.205

Table 3: Correlation of Δ with model capacity measures. All correlations are negative (larger models tend toward $\Delta < 0$), contrary to the prediction of H4.

All correlations are negative: larger models marginally *favour* reversed sequences. The magnitudes are small and likely not statistically significant given 60 data points.

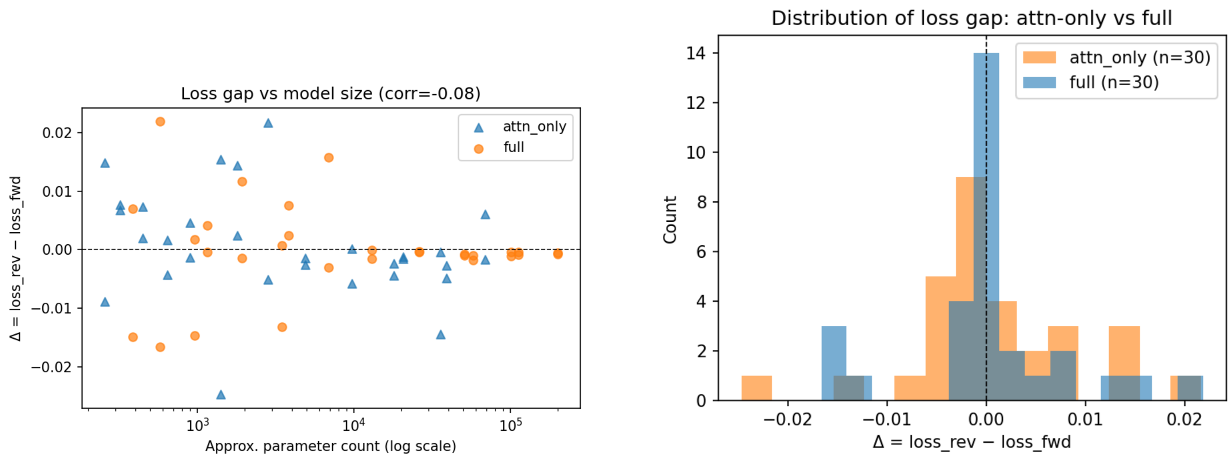


(a) Forward vs. reversed validation loss for all 60 matched configurations, coloured by n_{embd}. Points on the diagonal indicate identical performance; points above the diagonal indicate reversed sequences are harder. The data cluster tightly around the diagonal.



(b) $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ for each of the 60 configurations, sorted by magnitude. Most values cluster within ± 0.01 nats of zero. The extreme outlier ($\Delta \approx -0.025$) corresponds to the attention-only model `L4_d8_attn_noLN`.

Figure 1: Forward versus reversed validation loss across the full architecture sweep. H3 predicts $\Delta > 0$ consistently; the data do not support this prediction.



(a) Δ versus approximate parameter count. A slight downward trend indicates that larger models tend toward $\Delta \leq 0$, opposite to the prediction that larger models should *close* the gap.

(b) Distribution of Δ split by architecture type (attention-only vs. full transformer with MLP layers). Both distributions are centred near zero.

Figure 2: Dependence of the forward–reverse gap Δ on model capacity and architecture type.

6.3 Architectural Breakdown

Figure 3 shows the mean validation loss by n_{layer} and n_{embd} for both conditions.

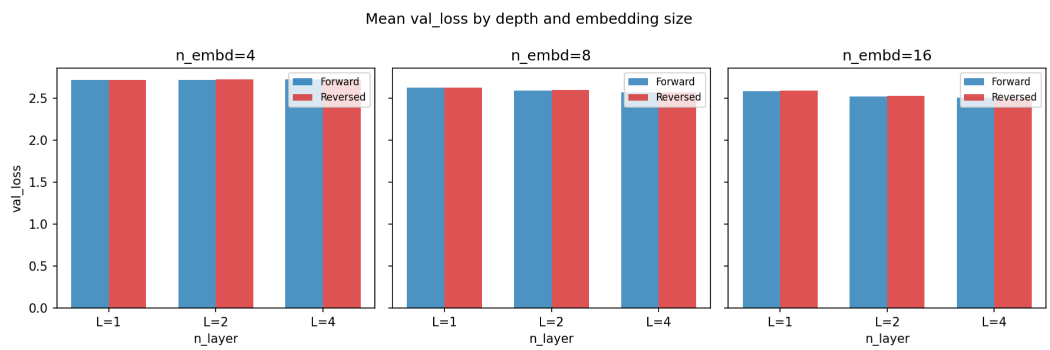


Figure 3: Mean validation loss grouped by n_{layer} and n_{embd}, for forward (blue) and reversed (orange) conditions. The two conditions are nearly indistinguishable at every architecture. The largest separations occur at small embedding dimension ($n_{\text{embd}} = 4$).

Notable patterns in the per-configuration breakdown:

- **Extreme outlier:** `L4_d8_attn_only_noLN` achieves $\Delta = -0.025$ nats (reversed is *easier*). This is the largest deviation in either direction.
- **Small models** ($n_{\text{embd}} = 4$) tend toward positive Δ, consistent with H3.
- **Large full models** ($n_{\text{embd}} \geq 32$, full transformer with MLP layers) mostly have $\Delta \leq 0$.

6.4 Interpretation

The near-zero mean gap suggests that at context length $T = 16$, the forward and reverse belief states are both sufficiently converged that there is negligible practical difference in prediction difficulty. Two non-exclusive interpretations are:

1. **Context window is sufficient.** The mixing time of the reverse chain at $k = 16$ may be short enough that $\Delta_{\text{Bayes}}(16) \approx 0$ for both directions. If the Bayes-optimal gap itself vanishes at $k = 16$, no transformer can show a gap either.

2. **Transformers do not implement Bayesian filtering.** The transformer may use a prediction strategy that is less sensitive to belief-state convergence rates, making the theoretical gap irrelevant to its empirical performance.

A direct computation of $\Delta_{\text{Bayes}}(16)$ with 50,000 Monte Carlo samples would distinguish these two explanations.

7 Activation Ablation Experiments

7.1 Overview

A complementary set of experiments investigated whether the belief-state geometry learned by the transformer is spatially localised within the residual stream. Using the best trained model ($L = 4$, $d = 8$, $H = 1$, no LayerNorm), we identified the low-dimensional subspace of the residual stream most aligned with the mixed-state (belief-state) representation via PCA regression on the activations at the final sequence position.

7.2 Direction Identification

The "fractal directions" are the principal components of the residual-stream activations that best predict the mixed-state representation (the belief distribution over hidden states) at the final token position. The orthogonal complement forms the "non-fractal" subspace. The leading two principal components account for the bulk of the explained variance, consistent with the low-dimensional fractal geometry of the HMM belief simplex.

7.3 Ablation Protocol

At inference time the activation at one or more sequence positions is replaced by its projection onto the *orthogonal complement* of the fractal subspace—i.e. the fractal-direction component is zeroed out. The model is then run to completion and the resulting cross-entropy loss is recorded. Two variants were tested:

1. **Last-position ablation:** Only the activation at the final sequence position is intervened upon, consistent with conventional activation-patching practice.

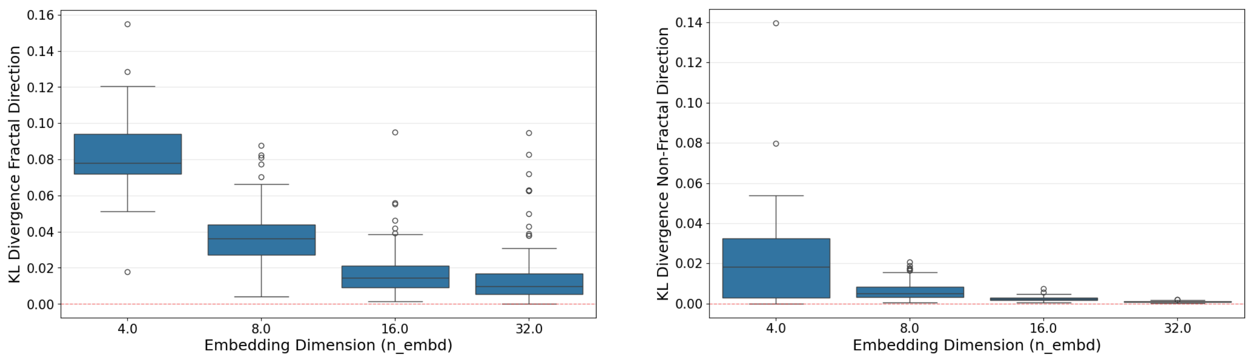
2. **Full-sequence ablation:** Every position's activation is replaced, preventing any accumulation of belief-state information along the entire sequence.

7.4 Results

The key finding is that **the fractal directions are substantially more important than the orthogonal subspace**:

- Ablating the fractal-direction component (zeroing the informative subspace) causes a much larger increase in cross-entropy than ablating the orthogonal complement. This asymmetry holds for both the last-position and full-sequence modes.
- Nevertheless, ablating the fractal direction does not entirely eliminate predictive performance, particularly for models with larger embedding dimensions. This suggests that larger models distribute information across a wider subspace.

Figures 4 and 5 show the last-position ablation results; Figure 6 shows the full-sequence ablation.



(a) Validation loss after zeroing the fractal-direction component at the last position, compared with zeroing the orthogonal complement (model variant A).

(b) Same as (a) for model variant B. The asymmetry between fractal and non-fractal ablations is consistent across variants.

Figure 4: Last-position ablation: cross-entropy loss when zeroing the fractal-direction or orthogonal-complement component of the residual stream at the final sequence position. Ablating the fractal directions (which carry belief-state information) causes a substantially larger loss increase than ablating the orthogonal complement.

These results confirm that the transformer compresses its belief-state representation into a low-dimensional subspace of the residual stream, consistent with the *mixed-state presentation* picture of HMM belief states.

8 Embedding Geometry

8.1 Cosine Similarity Structure

Analysis of the learned token-embedding matrix $W_E \in \mathbb{R}^{48\times d}$ reveals a strong 3×3 block structure in the cosine-similarity matrix: tokens within the same depth cluster are mapped to similar embedding directions, while cross-cluster pairs are near-orthogonal or anti-correlated. This is described in detail in `notes/model_comparison/cylinder_hmm_model_comparison.tex`; we summarise the key additional findings below.

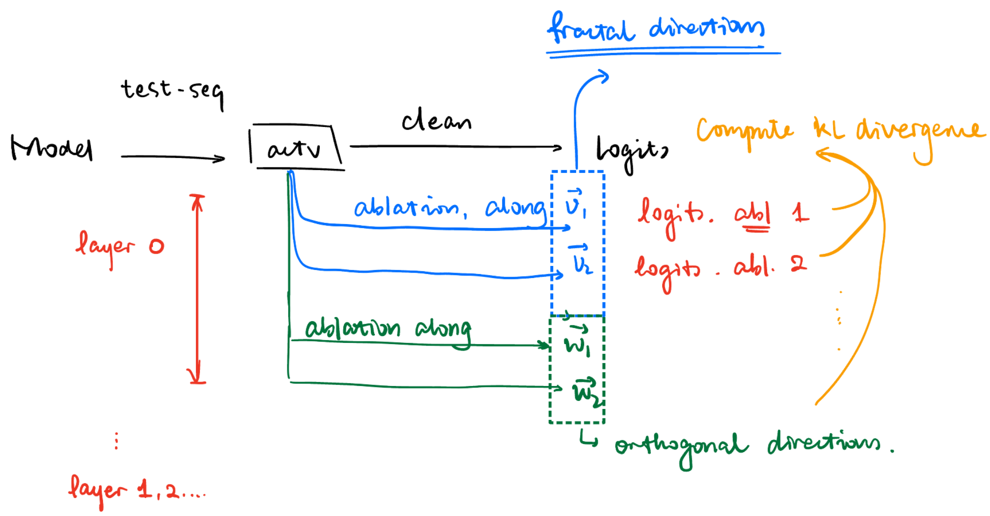
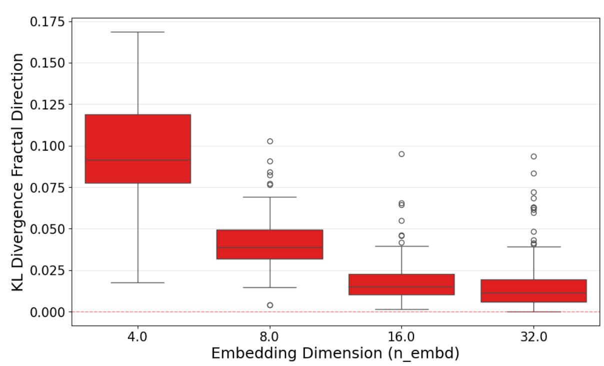
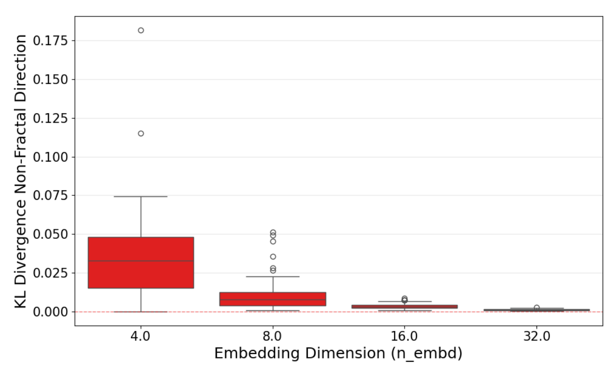


Figure 5: Summary of last-position ablation results across model sizes and configurations. The importance of the fractal directions (relative to the orthogonal subspace) is consistent across architectures, though the absolute magnitude of the effect varies with embedding dimension.



(a) Full-sequence ablation for model variant A: every position's fractal-direction component is zeroed, preventing any accumulation of belief-state information.



(b) Full-sequence ablation for model variant B. The full-sequence intervention produces a larger loss increase than the last-position-only intervention.

Figure 6: Full-sequence ablation: the fractal-direction component is zeroed at *every* sequence position, fully blocking belief-state accumulation throughout the context window. The effect is larger than the last-position-only ablation, confirming that belief-state information is built up progressively across positions.

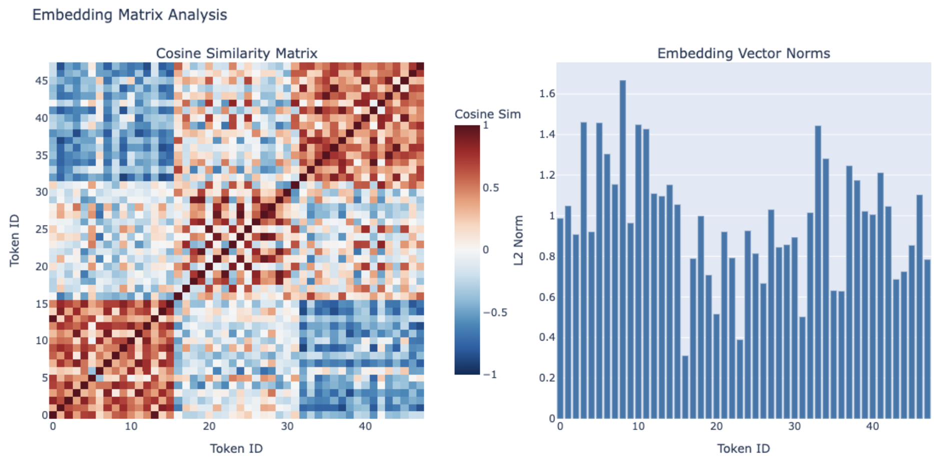


Figure 7: Cosine-similarity matrix of the learned token embeddings W_E. The 48 tokens are ordered by depth cluster (clusters of 16), revealing a clear 3×3 block structure: within-cluster cosine similarities are substantially higher than cross-cluster similarities. This mirrors the depth-level organisation of the Cylinder Graph HMM.

8.2 Norm Structure

The ℓ_2 norms of individual token embeddings do not correlate with the empirical token frequencies in the training data. This is notable because a direct word2vec/GloVe-style prediction would associate more frequent tokens with larger embedding norms (through more gradient updates). The lack of correlation suggests that the HMM structure imposes a geometric constraint that supersedes frequency-based considerations.

8.3 Relationship to PMI Embeddings

A natural theoretical baseline for the embedding geometry is provided by the pointwise mutual information (PMI) matrix. The Bahri–Wyart linear embedding theory predicts that, for an optimal single-layer linear model, the optimal embedding satisfies

$$W_E^\top W_E \;=\; \log \mathrm{PMI}, \tag{5}$$

where $\mathrm{PMI}(i,j) = \log[P(i,j)/(P(i)\,P(j))]$ is computed from the token co-occurrence statistics. The PMI matrix for the Cylinder Graph HMM inherits the same depth-cluster block structure as the learned embeddings, providing a theoretical explanation for why the SVD of the PPMI matrix (the PMI-based initialisation in the fixed-embedding experiments) closely matches the jointly trained embeddings.

9 Discussion and Open Questions

9.1 Summary of Findings

1. **H1 (entropy rate invariance) is confirmed.** Forward and reverse entropy rates agree to machine precision, as required by the stationarity of the process.

2. **H2 (finite-context asymmetry) has theoretical support.** Preliminary calculations show that the belief state converges ~ 2–$3\times$ more slowly for the reverse process at each

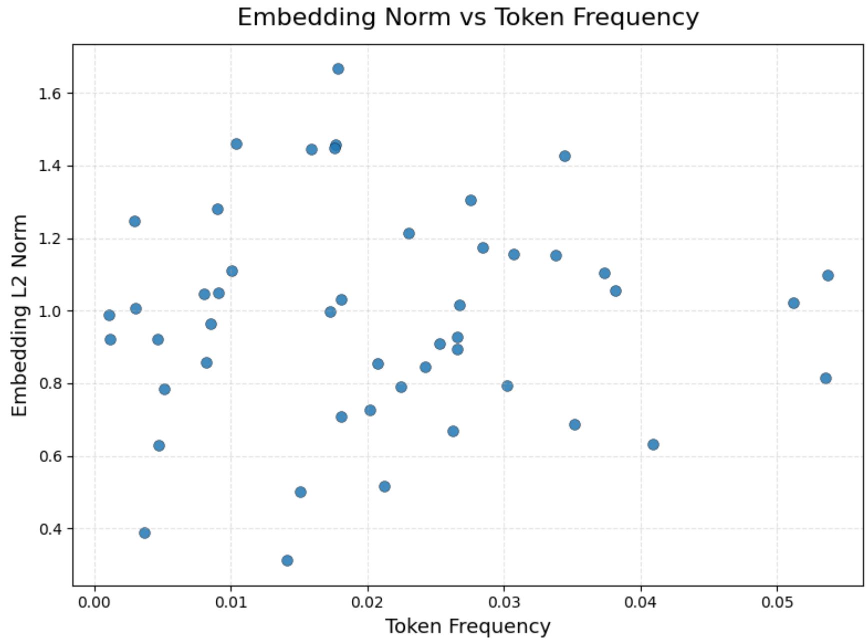


Figure 8: ℓ_2 norms of the 48 learned token embeddings, ordered by depth cluster. Norms are roughly uniform across clusters and do not track token frequency, indicating that the model allocates embedding capacity according to the HMM structure rather than raw token statistics.

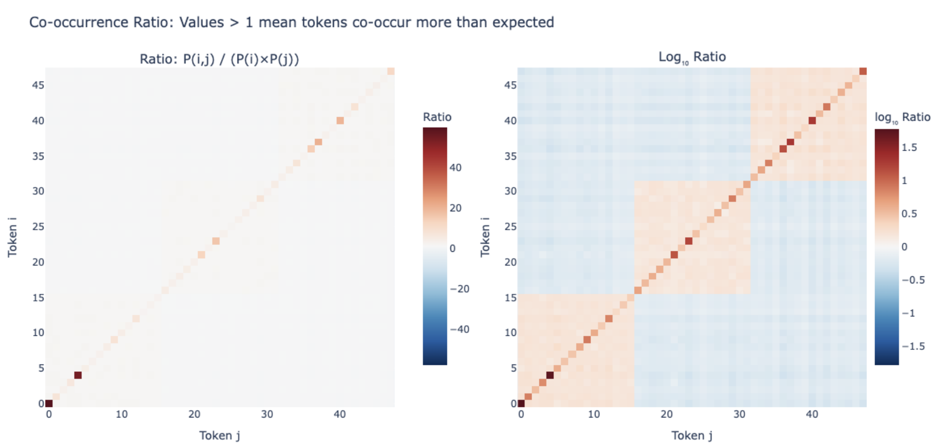


Figure 9: Comparison of the learned embedding Gram matrix $W_E^\top W_E$ with the log-PMI matrix computed from token co-occurrence statistics. The strong agreement between the two matrices supports the theoretical prediction $W_E^\top W_E \approx \log \mathrm{PMI}$ and explains why PMI-based (PPMI SVD) embedding initialisations perform well in the fixed-embedding experiments.

context position, implying $\Delta_{\text{Bayes}}(k) > 0$ for small k. Whether this gap survives at $k = 16$ is not yet established.

3. **H3 (transformer learnability gap) is not supported.** The mean forward–reverse loss difference is $+0.0002 \pm 0.0086$ nats across 60 configurations, consistent with zero.

4. **H4 (architecture dependence) shows a weak negative correlation** between model size and Δ, opposite to the prediction.

5. **Belief-state geometry is spatially localised.** Activation ablations confirm that the model stores its belief-state representation in a low-dimensional subspace of the residual stream, with the fractal directions carrying substantially more predictive information than the orthogonal complement.

6. **Embedding geometry is consistent with PMI structure.** The learned embeddings satisfy an approximate $W_E^\top W_E \approx \log \text{PMI}$ relationship, connecting the empirical geometry to the theoretical optimum for linear models.

9.2 Open Questions

- **Bayes-optimal gap at $k = 16$:** Run `src/reverse_hmm_analysis.py` with 50,000 samples and `--max_context 16`. If $\Delta_{\text{Bayes}}(16) \approx 0$, the near-zero transformer gap is fully explained by the HMM geometry.

- **N-gram baselines on reversed data:** Do n-gram models also show no forward–reverse gap? If they do, the near-zero Δ is a general property of any fixed-context predictor and not specific to transformers.

- **Outlier investigation:** The `L4_d8_attn_only_noLN` configuration achieves $\Delta = -0.025$ nats. What mechanism makes an attention-only model find reversed sequences *easier*?

- **Longer context:** Training on sequences of length $T > 16$ would test whether the near-zero gap persists or whether a gap emerges when the context window exceeds the mixing time of the reverse chain.

- **Ablations at full scale:** The activation ablations were performed on a single model. Repeating across the full architecture sweep would determine whether fractal-direction localisation is universal or architecture-dependent.

- **Fractal directions in the reverse model:** Does the reverse-trained transformer develop the same low-dimensional belief-state geometry? If so, what is the relationship between the fractal subspaces of the forward and reverse models?